

DEGREE PRESSURE, MULTIFRACTAL FIBRES, AND PROFILE RIGIDITY FOR FULL ZIP SHIFTS

ANONYMOUS

ABSTRACT. Let $\tau : S \rightarrow Z$ be a surjection of finite alphabets and let F_τ be the associated full zip shift. Its local degree is $d_\tau(x) = |F_\tau^{-1}(x)| = k_{x_{-1}}$, where $k_z = |\tau^{-1}(z)|$. We study the intrinsic potential $\phi_\tau = \log d_\tau$ for a nonuniform fibre profile. Its pressure, equilibrium state, and curvature are explicit:

$$P_\tau(t) = \log \sum_{z \in Z} k_z^{t+1}, \quad p_t(s) = \frac{k_{\tau(s)}^t}{\sum_z k_z^{t+1}}, \quad P_\tau''(t) = \text{Var}_{r_t}(\log k_z).$$

We compute the Bowen entropy of every degree-exponent level set. For $\alpha \in [\log k_{\min}, \log k_{\max}]$ it equals

$$r: \sum \max_{r_z \log k_z = \alpha} (H(r) + \alpha) = \inf_{t \in \mathbb{R}} (P_\tau(t) - t\alpha),$$

and the level set is empty outside this interval.

The full pressure curve is also a complete invariant inside this family. Two full zip shifts are topologically conjugate if and only if their fibre-size multisets agree, equivalently if and only if their degree-pressure curves agree for all real t . A degree-weighted periodic identity provides a finite-orbit form of the same exponential sum. Existing metric- and folding-entropy formulae for extended shifts are used with explicit attribution; at the equilibrium state they yield $h_{\mu_t} = P_\tau(t) - tP_\tau'(t)$ and folding entropy $\mathcal{F}_{\mu_t} = P_\tau'(t)$.

1. INTRODUCTION

A zip shift records future symbols in one alphabet and past symbols in a quotient alphabet. When a future symbol crosses the origin, a surjection forgets which element of its fibre was present. This gives a compact symbolic model of a finite-to-one noninvertible map. The formal zip-shift space, its sliding block codes, local homeomorphism structure, and periodic points are developed by Lamei and Mehdipour [3]. The same system appears under the name *extended shift*; Martins et al. [4] compute metric and folding entropies for its Bernoulli measures. Those results are owner-subtracted here.

For a uniform n -to-one zip shift the local degree is constant, so its logarithm carries no fluctuations. Uniform systems are treated through square entropy and intrinsic ergodicity by Mehdipour and Shaladehi [6]. A nonuniform fibre profile changes the question: the orbit sees a sequence of local degrees, and the exponential growth of their products depends on how frequently the future visits each fibre. The intrinsic potential

$$\phi_\tau(x) = \log |F_\tau^{-1}(x)|$$

turns that loss of information into an additive observable.

This paper gives one closed theorem package for that observable.

- (1) The pressure and its unique equilibrium states satisfy

$$P_\tau(t) = \log \sum_{z \in Z} k_z^{t+1}, \quad p_t(s) \propto k_{\tau(s)}^t.$$

The derivative is the equilibrium average of $\log d_\tau$, and the second derivative detects whether the profile is nonuniform.

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- (2) The degree-exponent level sets have an exact Bowen-entropy spectrum, both as a constrained Shannon maximum and as the Legendre transform of P_τ .
- (3) The entire pressure curve recovers the fibre-size multiset. Combined with an explicit one-block construction, this yields the equivalence among topological conjugacy, equal fibre profiles, and equal degree-pressure curves.
- (4) Weighted periodic points satisfy

$$\sum_{x \in \text{Fix}(F_\tau^n)} \exp \left(t \sum_{j=0}^{n-1} \phi_\tau(F_\tau^j x) \right) = \left(\sum_z k_z^{t+1} \right)^n.$$

The natural extension of a full zip shift is the ordinary full two-sided $|S|$ -shift. We prove the explicit identification because it is the shortest route to pressure, but we use it only as infrastructure. Likewise, the metric- and folding-entropy formulae themselves remain those of Martins et al. [4]; our statement substitutes the equilibrium weights and connects the resulting quantities to P'_τ .

Bowen's entropy for noncompact sets provides the level-set notion [2]; digit-frequency multifractal arguments give adjacent context [1]. Here the proof reduces directly to types on the future alphabet and keeps the multiplicity of every fibre.

Section 2 defines the map and its extension. Pressure and entropy identities are proved in section 3. Periodic data and conjugacy rigidity appear in section 4, the multifractal spectrum in section 5, and source boundaries in section 7.

2. THE FULL ZIP SHIFT AND ITS NATURAL EXTENSION

Let S and Z be nonempty finite sets, and let $\tau : S \rightarrow Z$ be surjective. Set

$$X_\tau = Z^{\mathbb{Z}_{<0}} \times S^{\mathbb{Z}_{\geq 0}}.$$

We write a point as $x = (\dots, x_{-2}, x_{-1}; x_0, x_1, \dots)$ and define

$$(2.1) \quad F_\tau x = (\dots, x_{-2}, x_{-1}, \tau(x_0); x_1, x_2, \dots).$$

This is the full one-block zip shift of Laméi and Mehdipour [3], and the same formula defines the extended shift of Martins et al. [4]. For $z \in Z$, put

$$k_z = |\tau^{-1}(z)|, \quad m_k = |\{z \in Z : k_z = k\}|.$$

The multiset $\{k_z : z \in Z\}$ is the *fibre profile*.

Lemma 2.1 (Local degree). *The map F_τ is a local homeomorphism and*

$$(2.2) \quad d_\tau(x) := |F_\tau^{-1}(x)| = k_{x_{-1}}.$$

Thus $\phi_\tau = \log d_\tau$ is a locally constant intrinsic potential.

Proof. For a preimage y of x , all coordinates except y_0 are forced by (2.1): $y_j = x_{j-1}$ for $j \neq 0$. The remaining condition is $\tau(y_0) = x_{-1}$, giving exactly $k_{x_{-1}}$ choices. On the cylinder where $x_{-1} = z$, the inverse branches obtained by choosing $s \in \tau^{-1}(z)$ are continuous. $\square$

Let $\sigma : S^{\mathbb{Z}} \rightarrow S^{\mathbb{Z}}$ be the ordinary left shift. Define

$$(2.3) \quad \pi : S^{\mathbb{Z}} \longrightarrow X_\tau, \quad \pi(t)_i = \begin{cases} \tau(t_i), & i < 0, \\ t_i, & i \geq 0. \end{cases}$$

Then $F_\tau \pi = \pi \sigma$.

Proposition 2.2 (Explicit natural extension). *The inverse-limit natural extension of (X_τ, F_τ) is conjugate to $(S^{\mathbb{Z}}, \sigma)$, with current-coordinate factor (2.3). Under this identification,*

$$(2.4) \quad \phi_\tau \circ \pi(t) = \log k_{\tau(t_{-1})}.$$

Invariant probabilities correspond affinely, and corresponding measures have equal metric entropy.

Proof. An inverse history is a sequence $(x^{(0)}, x^{(-1)}, \dots)$ satisfying $F_\tau x^{(-j)} = x^{(-j+1)}$. Recover $t_i = x_i^{(0)}$ for $i \geq 0$ and $t_{-j} = x_0^{(-j)}$ for $j \geq 1$. Conversely, a two-sided word t gives the history whose j th predecessor has future beginning at t_{-j} and whose past is the corresponding τ -image. These coordinate formulae are continuous and inverse to one another, and advancing the history is the left shift. Formula (2.4) follows from $\pi(t)_{-1} = \tau(t_{-1})$. The invariant-measure and entropy statements are the standard measure-theoretic properties of the natural extension. $\square$

In particular, the topological entropy is $\log |S|$. This ordinary fact is not the invariant used below: different fibre profiles can have the same $|S|$ and hence the same entropy.

3. DEGREE PRESSURE AND EQUILIBRIUM STATES

For $t \in \mathbb{R}$, let $P_\tau(t)$ denote the topological pressure of $t\phi_\tau$ for F_τ . Define

$$(3.1) \quad Q_\tau(t) = \sum_{z \in Z} k_z^{t+1} = \sum_{s \in S} k_{\tau(s)}^t.$$

Theorem 3.1 (Pressure, equilibrium, and curvature). *For every $t \in \mathbb{R}$,*

$$(3.2) \quad P_\tau(t) = \log Q_\tau(t).$$

There is a unique equilibrium state μ_t . Its natural extension is the Bernoulli measure on $S^{\mathbb{Z}}$ with one-symbol probabilities

$$(3.3) \quad p_t(s) = \frac{k_{\tau(s)}^t}{Q_\tau(t)}.$$

If

$$(3.4) \quad r_t(z) = \sum_{s \in \tau^{-1}(z)} p_t(s) = \frac{k_z^{t+1}}{Q_\tau(t)},$$

then

$$(3.5) \quad P'_\tau(t) = \sum_z r_t(z) \log k_z, \quad P''_\tau(t) = \text{Var}_{r_t}(\log k_z).$$

Hence P_τ is strictly convex exactly when the fibre profile is nonuniform.

Proof. By [proposition 2.2](#), the variational functional for F_τ equals that for the full shift on $S^{\mathbb{Z}}$ with the one-coordinate potential $t \log k_{\tau(s)}$. For any shift-invariant measure ν , its entropy rate is bounded by the Shannon entropy $H(p)$ of its one-symbol marginal p . The finite-alphabet Gibbs inequality gives

$$(3.6) \quad H(p) + t \sum_{s \in S} p(s) \log k_{\tau(s)} \leq \log \sum_{s \in S} k_{\tau(s)}^t,$$

with equality only at (3.3); entropy-rate equality then forces the Bernoulli process. This proves (3.2) and uniqueness.

Summing (3.3) over a fibre gives (3.4). Differentiating the finite exponential sum yields (3.5). Its variance vanishes for one, hence every, t exactly when all values $\log k_z$ coincide. $\square$

Two parameter values have immediate meanings:

$$(3.7) \quad P_\tau(0) = \log |S|, \quad P_\tau(-1) = \log |Z|.$$

Thus the curve contains both alphabet sizes, while its curvature records the nonuniform distribution of information loss.

The next corollary explicitly uses the main entropy formulae of Martins et al. [4]; it is not a reattribution of those formulae.

Corollary 3.2 (Metric and folding entropy along equilibrium). *For the equilibrium state μ_t ,*

$$(3.8) \quad h_{\mu_t}(F_\tau) = P_\tau(t) - tP'_\tau(t), \quad \mathcal{F}_{\mu_t}(F_\tau) = P'_\tau(t).$$

Proof. Theorem A of Martins et al. [4] identifies metric entropy with $H(p_t)$. From (3.3), $-\log p_t(s) = P_\tau(t) - t \log k_{\tau(s)}$, so averaging gives the first identity. Their Theorem B writes folding entropy as the average conditional entropy inside the fibres. Equation (3.3) is uniform within each fibre, so the conditional entropy over z is $\log k_z$. Averaging with r_t and using (3.5) gives the second identity. $\square$

4. WEIGHTED PERIODIC DATA AND PROFILE RIGIDITY

The same exponential sum in (3.1) appears without a variational argument when periodic points are weighted by their local degrees.

Proposition 4.1 (Degree-weighted periodic identity). *For every $n \geq 1$ and $t \in \mathbb{R}$,*

$$(4.1) \quad \sum_{x \in \text{Fix}(F_\tau^n)} \exp \left(t \sum_{j=0}^{n-1} \phi_\tau(F_\tau^j x) \right) = Q_\tau(t)^n.$$

In particular $|\text{Fix}(F_\tau^n)| = |S|^n$.

Proof. Each word $(s_0, \dots, s_{n-1}) \in S^n$ determines one fixed point of F_τ^n : repeat the word on nonnegative coordinates and repeat its τ -image on negative coordinates. Every fixed point arises uniquely in this way. Along its orbit the n local degrees are the cyclic list $k_{\tau(s_0)}, \dots, k_{\tau(s_{n-1})}$. Summing the product of their t th powers over S^n factorizes as $(\sum_s k_{\tau(s)}^t)^n = Q_\tau(t)^n$. $\square$

The weighted orbit sums also give a closed zeta function. For $|u| < Q_\tau(t)^{-1}$, set

$$\zeta_\tau(t, u) = \exp \left(\sum_{n \geq 1} \frac{u^n}{n} \sum_{x \in \text{Fix}(F_\tau^n)} e^{t S_n \phi_\tau(x)} \right).$$

Corollary 4.2 (Degree-weighted zeta function). *For every real t ,*

$$(4.2) \quad \zeta_\tau(t, u) = \frac{1}{1 - u Q_\tau(t)}.$$

Its smallest positive pole is $e^{-P_\tau(t)}$.

Proof. Insert (4.1) and use $\sum_{n \geq 1} (uQ)^n/n = -\log(1 - uQ)$. $\square$

We now compare two surjections $\tau : S \rightarrow Z$ and $\kappa : T \rightarrow W$. A topological conjugacy means a homeomorphism $H : X_\tau \rightarrow X_\kappa$ satisfying $H F_\tau = F_\kappa H$.

Theorem 4.3 (Profile, pressure, and conjugacy). *The following are equivalent.*

- (1) (X_τ, F_τ) and (X_κ, F_κ) are topologically conjugate.
- (2) The fibre-size multisets $\{k_z : z \in Z\}$ and $\{|\kappa^{-1}(w)| : w \in W\}$ agree.
- (3) $P_\tau(t) = P_\kappa(t)$ for every $t \in \mathbb{R}$.

Equal profiles admit a one-block conjugacy on the negative and nonnegative coordinates separately.

Proof. Conjugacy preserves local degree pointwise because it bijects the preimage sets of corresponding points. It also bijects fixed points. There is one fixed point for each $s \in S$, and that point has degree $k_{\tau(s)}$. Consequently the number of fixed points of degree k is

$$(4.3) \quad N_k = k m_k.$$

The conjugacy-invariant histogram N_k recovers every multiplicity $m_k = N_k/k$, proving (1) $\Rightarrow$ (2).

If the profiles agree, choose a bijection $\beta : Z \rightarrow W$ matching equal-size fibres and, for each z , a bijection $\alpha_z : \tau^{-1}(z) \rightarrow \kappa^{-1}(\beta z)$. Put $\alpha(s) = \alpha_{\tau(s)}(s)$ and define

$$(4.4) \quad H(x)_i = \begin{cases} \beta(x_i), & i < 0, \\ \alpha(x_i), & i \geq 0. \end{cases}$$

The relation $\kappa\alpha = \beta\tau$ shows directly from (2.1) that $HF_\tau = F_\kappa H$. Coordinatewise inverse bijections make H a conjugacy, proving (2) $\Rightarrow$ (1).

Equal profiles give equal sums in (3.1), hence (2) $\Rightarrow$ (3). Conversely suppose the pressure curves agree and set

$$(4.5) \quad R_\tau(u) = \exp P_\tau(u-1) = \sum_{k \geq 1} m_k k^u.$$

The largest fibre size is $K = \lim_{u \rightarrow \infty} R_\tau(u)^{1/u}$, and its multiplicity is $m_K = \lim_{u \rightarrow \infty} R_\tau(u)/K^u$. Subtract $m_K K^u$ and repeat on the finite remaining sum. This recovers the whole profile from the curve and proves (3) $\Rightarrow$ (2). $\square$

The fixed-point argument already proves profile necessity. The pressure formulation is stronger as a thermodynamic representation: one convex function simultaneously carries the profile, equilibrium degree averages, and the spectrum proved next.

5. THE DEGREE-EXPONENT SPECTRUM

For $x \in X_\tau$, define the forward degree exponent when the limit exists:

$$(5.1) \quad \lambda(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \prod_{j=0}^{n-1} d_\tau(F_\tau^j x), \quad E_\tau(\alpha) = \{x : \lambda(x) = \alpha\}.$$

We use Bowen's topological entropy h_B for a possibly noncompact level set [2].

Lemma 5.1 (Future-symbol reduction). *For every x and $n \geq 2$,*

$$(5.2) \quad \sum_{j=0}^{n-1} \phi_\tau(F_\tau^j x) = \log k_{x_{-1}} + \sum_{i=0}^{n-2} \log k_{\tau(x_i)}.$$

The first term is uniformly bounded. Moreover, the negative coordinates contribute zero exponential growth to forward Bowen covers. Hence the Bowen entropy of $E_\tau(\alpha)$ equals the digit-average level-set entropy in the one-sided full shift on S for the weights $a_s = \log k_{\tau(s)}$.

Proof. The degree at time zero is $k_{x_{-1}}$. For $j \geq 1$, the symbol at coordinate -1 of $F_\tau^j x$ is $\tau(x_{j-1})$, giving (5.2). Since the alphabet is finite, the boundary term divided by n tends uniformly to zero. For a fixed metric scale, an n -step Bowen name fixes $n + O(1)$ future symbols but only $O(1)$ initial past symbols. Those past choices multiply cover counts by a factor independent of n and therefore do not change entropy. $\square$

Let $\Delta(S)$ and $\Delta(Z)$ denote the corresponding probability simplices, and write H for Shannon entropy with natural logarithms.

Theorem 5.2 (Exact multifractal spectrum). *Put $k_{\min} = \min_z k_z$ and $k_{\max} = \max_z k_z$. If $\alpha \notin [\log k_{\min}, \log k_{\max}]$, then $E_\tau(\alpha) = \emptyset$. For every α in this interval,*

$$(5.3) \quad h_B(E_\tau(\alpha)) = \max_{p \in \Delta(S): \sum_s p_s \log k_{\tau(s)} = \alpha} H(p)$$

$$(5.4) \quad = \max_{r \in \Delta(Z): \sum_z r_z \log k_z = \alpha} (H(r) + \alpha)$$

$$(5.5) \quad = \inf_{t \in \mathbb{R}} (P_\tau(t) - t\alpha).$$

For an interior α and a nonuniform profile, the infimum is attained at the unique t satisfying $P'_\tau(t) = \alpha$.

Proof. By lemma 5.1, it suffices to count words in S^n with empirical a_s -average near α . For a type p , the number of such words is $\exp(nH(p) + o(n))$, uniformly over the polynomially many types. Covering all types whose averages lie in a shrinking neighbourhood of α gives the upper bound in (5.3). Conversely, for any p satisfying the constraint, the set of p -generic sequences has Bowen entropy $H(p)$: the usual typical-word separated sets give the lower bound, or

equivalently one applies the entropy distribution argument to the Bernoulli p measure. Taking the maximum proves (5.3). It also shows that the level set is empty outside the convex hull of the weights, which is the stated interval.

Given p , group it by fibres: $r_z = \sum_{s \in \tau^{-1}(z)} p_s$. The entropy chain rule gives

$$(5.6) \quad H(p) = H(r) + \sum_z r_z H(p(\cdot | z)) \leq H(r) + \sum_z r_z \log k_z = H(r) + \alpha.$$

Equality holds by distributing mass uniformly within each fibre. This proves (5.4).

For any t and any feasible p , the Gibbs variational inequality (3.6) rearranges to $H(p) \leq P_\tau(t) - t\alpha$. Hence the maximum is at most the infimum in (5.5). If α is interior, continuity and strict monotonicity of P'_τ give a parameter t with $P'_\tau(t) = \alpha$; the distribution p_t is feasible and equality in (3.6) proves the reverse inequality. At the endpoints, take $t \rightarrow -\infty$ or $t \rightarrow +\infty$ and use continuity of the constrained maximum. $\square$

At an endpoint with fibre size k , the formula retains multiplicity:

$$h_B(E_\tau(\log k)) = \log(km_k)$$

when k is the unique extremal size value (all fibres of that size are included). The spectrum reaches the full entropy $\log |S|$ at

$$\alpha = P'_\tau(0) = \sum_z \frac{k_z}{|S|} \log k_z.$$

Together with corollary 3.2, the tangent parameter describes both the folding entropy of μ_t and the degree exponent of its generic points.

6. TWO FOUR-SYMBOL CONTROLS

The ordinary topological entropy cannot distinguish fibre profiles with the same forward alphabet size. Consider two surjections with $|S| = 4$:

$$\mathbf{k} = (1, 3), \quad \tilde{\mathbf{k}} = (2, 2).$$

Both full zip shifts have topological entropy $\log 4$, but

$$(6.1) \quad P_{(1,3)}(t) = \log(1 + 3^{t+1}), \quad P_{(2,2)}(t) = (t + 2) \log 2.$$

The first curve is strictly convex, while the second is affine. Theorem 4.3 therefore separates the systems even though their natural extensions are both the full four-shift.

For the nonuniform profile $(1, 3)$, write $\theta = \alpha / \log 3$. Formula (5.4) becomes

$$(6.2) \quad h_B(E(\alpha)) = -\theta \log \theta - (1 - \theta) \log(1 - \theta) + \alpha, \quad 0 \leq \alpha \leq \log 3.$$

At $\alpha = 0$ the level set uses the single symbol over the one-element fibre and has entropy zero. At $\alpha = \log 3$ it uses the three symbols over the three-element fibre and has entropy $\log 3$. The maximum $\log 4$ occurs at $\theta = 3/4$, the fibre frequency induced by the uniform distribution on the four future symbols.

For the uniform profile $(2, 2)$, every point has degree exponent $\log 2$. The spectrum consists of one level set of entropy $\log 4$, matching the affine pressure and zero curvature. This is the degenerate case adjacent to the uniform n -to-one theory of Mehdipour and Shaldehi [6].

7. SOURCE BOUNDARY AND LIMITATIONS

The formal zip-shift definitions, local homeomorphism results, sliding block codes, and periodic-point setting belong to Lamei and Mehdipour [3]. Uniform n -to-one intrinsic ergodicity and square-entropy classification are treated by Mehdipour and Shaldehi [6]. Most directly, Martins et al. [4] study exactly the map (2.1) under the name extended shift and prove the Bernoulli metric- and folding-entropy formulae used in corollary 3.2. We do not present the natural extension, ordinary entropy, local degree, or their Theorems A–B as isolated paper claims.

TABLE 1. What ordinary entropy forgets and degree pressure retains for two full zip shifts with $|S| = 4$.

Quantity	profile (1, 3)	profile (2, 2)
Natural extension	full four-shift	full four-shift
h_{top}	$\log 4$	$\log 4$
degree range	$\{1, 3\}$	$\{2\}$
$P(t)$	$\log(1 + 3^{t+1})$	$(t + 2) \log 2$
$P''(t)$	positive	zero
degree-exponent interval	$[0, \log 3]$	$\{\log 2\}$

Recent adjacent directions include zip cellular automata [7] and symbolic encodings of finite-to-one local homeomorphisms [5]. These works motivate the system class but do not alter the scope of the proofs here. The manuscript’s residual mathematical package is the intrinsic degree pressure, its equilibrium and curvature, the full degree-exponent spectrum, and the equivalence between pressure-profile recovery and topological conjugacy.

Three restrictions are important. First, the space is full: future symbols have no transition constraints. For a proper zip subshift, pressure is controlled by a weighted transition operator rather than the scalar sum $Q_\tau(t)$. Second, the conjugacy classification is internal to full one-block zip shifts; it does not classify arbitrary finite-to-one local homeomorphisms. Third, the bounded source search found no primary article stating the exact pressure/spectrum/profile package, but such a search is not a priority certificate. External release therefore remains on hold for a specialist audit, including current work on thermodynamic formalism for zip maps.

The companion script exhausts periodic words through length five for several integer weights, checks profile recovery on finite examples, differentiates the displayed pressure numerically against its exact mean/variance formulae, and verifies the binary spectrum’s entropy maximum. These are regression checks only; all formulae are proved symbolically in the manuscript.

8. CONCLUSION

For a nonuniform full zip shift, local degree is not merely a pointwise topological invariant. Its logarithm generates a pressure curve that simultaneously determines equilibrium measures, folding averages, degree fluctuations, weighted periodic data, and the Bowen-entropy spectrum of orbit exponents. The finite exponential sum behind that curve also recovers every fibre multiplicity, yielding a complete conjugacy invariant within the full one-block family.

The full-system assumption makes the pressure scalar and the spectrum a finite-dimensional entropy maximization. A next step is to replace the full future shift by a mixing SFT and determine which parts of profile recovery survive when the degree potential interacts with transition constraints. That extension is outside the present theorem contract.

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